

Graph neural network for 3-dimensional structures including dihedral angles for molecular property prediction

Sri Abhirath Reddy Sangala¹, Shampa Raghunathan¹

1. Mahindra University

Abstract

The prediction of molecular properties using graph neural network (GNN) based approaches has attracted great attention in recent years. Topological molecular graphs are commonly used for representing molecules in machine learning (ML). However, the challenge is to utilize the complete geometry information, like, bonds, angles and dihedral angles while processing a molecular graph. In this work we present predictive GNN accounting three-dimensional molecular structures including the dihedral angles (GNN3Dihed) in a systematic manner. Additionally, we demonstrate that the usage of autoencoders to generate latent space embeddings for usually sparse atomic and bond vectors reduces the number of parameters in the message passing stage while not reducing performance. We compare the performance of GNN3Dihed with state-of-the-art baselines on several tasks (regression and classification), e.g., solubility prediction, toxicity prediction, binding affinity, and quantum mechanical property prediction, and showed that the present architecture often outperforms other models—demonstrating the importance of 3D structural information for ML in chemistry.

Keywords

Molecular property prediction, Graph neural network, Three-dimensional chemical structure, Solubility, Toxicity, Dipole moments

Graph neural network for 3-dimensional structures including dihedral angles for molecular property prediction

Sri Abhirath Reddy Sangala^a and Shampa Raghunathan^{*a}

^aÉcole Centrale School of Engineering, Mahindra University, Hyderabad 500043, India
shampa.raghunathan@mahindrauniversity.edu.in

Abstract

The prediction of molecular properties using graph neural network (GNN) based approaches has attracted great attention in recent years. Topological molecular graphs are commonly used for representing molecules in machine learning (ML). However, the challenge is to utilize the complete geometry information, like, bonds, angles and dihedral angles while processing a molecular graph. In this work we present predictive GNN accounting three-dimensional molecular structures including the dihedral angles (GNN3Dihed) in a systematic manner. Additionally, we demonstrate that the usage of autoencoders to generate latent space embeddings for usually sparse atomic and bond vectors reduces the number of parameters in the message passing stage while not reducing performance. We compare the performance of GNN3Dihed with state-of-the-art baselines on several tasks (regression and classification), *e.g.*, solubility prediction, toxicity prediction, binding affinity, and quantum mechanical property prediction, and showed that the present architecture often outperforms other models—demonstrating the importance of 3D structural information for ML in chemistry.

1 Introduction

Machine learning (ML) has become an eminent research field not only in computer science but also in natural sciences due to tremendous progress in ML algorithms with the availability of high-performing computing facilities^[1–3]. Molecular property prediction which is an essential part of computational drug and materials discovery is now commonly done by using machine learning models^[4–6]. Most of these models rely on hand-engineered representation of molecules^[7–14]. However, topological molecular information for example, atoms and bonds can be represented as nodes and edges of a molecular graph, respectively^[15,16]. Molecular graphs subsequently can be processed by neural networks to predict molecular properties^[17–23]. A few of these graph-based neural networks formally known as graph neural networks (GNNs) additionally included angular information via atom pairs interactions, and was shown to reach better prediction accuracies^[24].

GNNs for molecular property prediction have become an emerging tool since message-passing neural network (MPNN) of Gilmer *et al.* was introduced^[18]. However, performance of such GNNs depend on the choice of architecture. Earlier approaches involved fingerprint-based^[11] or SMILES-based^[25,26] representations only to learn local atomic interactions via node-network, and relative distances via edge-network. State-of-the-art (SOTA) GNN/MPNN architectures encode molecular geometries using message passing principles, passing information

within a graph, utilizing transfer learning^[27]. A well-known list includes, DeepChem^[28], ChemProp^[29], SchNet^[30], PaiNN^[31], DimeNet^[32], GemNet^[33], MACE^[34], and Vanilla DisGNN^[35] etc..

One such model, widely used in the community is E(3)-invariant SchNet^[30] for learning quantum mechanical (QM) properties. For *updating* node states, this model uses *messages* informed by radial basis functions expanded in terms of inter-atomic distances between two nodes. Also, pair-wise distances, and angle between two edges are unchanged upon global rotations, translations, and reflections. To be SE(3)-invariant, 3D-GNNs consider torsional angle information too. One such model is GemNet^[33]. Very recently, E(3)-equivariant^[36] and SE(3)-equivariant^[37] GNNs have shown to predict tensorial properties such as multipole moments. Cremer *et al.* have reported equivariant GNN by considering three-dimensional (3D) geometry information for toxicity prediction. The GEM^[24] is one such model that includes atom, bond, and bond-angle graphs simultaneously, and shows efficient molecular representation learning is essential for molecular property prediction.

We provide a framework where the 3D geometry of a molecule, essentially atoms, bonds, angles and dihedral angles is represented using graphs. A graph containing the information regarding the angles is generated by considering the line graph^[39] of the primary topological graph of the molecule. Another secondary line graph of this bond-angle graph is generated to contain information regarding torsional angles. We describe a mechanism whereby information from message passing can be transferred between these graphs, allowing information regarding angles and torsional angles to flow to the primary graph. Additionally, we propose using latent space representations of atomic and bond vectors as opposed using the original vectors in message passings.

Our contributions towards building a predictive GNN accounting 3D molecular geometry, including dihedral angles (GNN3Dihed) can be summarized as follows: (a) propose an approach to process the full 3D information of a molecule (b) describe a mechanism via which information is transferred between graphs (c) propose the usage of latent space representations for atomic and bond vectors. The proposed model is applied on 12 regression datasets, and 10 classification datasets involving solubility prediction, toxicity prediction, binding affinity, and quantum mechanical property prediction. QM9iso—a subset of the original QM9 dataset consisting of 7430 constitutional isomers contains dipole moments (Debye unit) was specifically explored to assess contributions coming from multi-graph representation of a molecule. Extensive evaluation and comparisons suggest that GNN3Dihed is capable of giving some of the best predictions of molecular properties.

2 Methods

In this section, we introduce GNN3Dihed, outlining the key components and processes that define our model. We begin by discussing the data generation pipeline, which translates simplified molecular input line entry system (SMILES)^[25,26] strings into input representations. We highlight our novel application of autoencoders for producing atomic and bond vector representations. We then describe how this data is represented as a collection of graphs. Here we note a matter of terminology, we will often refer to this collection of graphs as a multi-graph (not the conventional graph theoretic meaning). Following that, we delve into the message-passing mechanism within a single graph and explain how information flows between graphs, enabling lower-level geometric details to influence higher-level molecular representations. Lastly, we discuss how this representation is used for classification or regression tasks.

2.1 Molecular Representation Generation

A primary goal of our work is to fully leverage the 3D geometric and topological information of the molecule. Using SMILES, we generate: (a) atomic vectors, (b) bond vectors, (c) angles, and (d) dihedral angles. We assert that the full 3D information is captured in the input by accounting for bond distances, angles, and dihedral angles. Additional atomic features (*e.g.*, atomic number, atomic weight) and bond characteristics (*e.g.*, single/double bond, ring membership) are also incorporated to form atomic and bond vectors. A complete list of features used to generate these vectors is available in Tables 1 and 2.

2.2 Autoencoders for Atomic/Bond Latent Space Representation

We utilize autoencoders to capture latent representations of the atomic and bond vectors, ensuring that all subsequent computations are carried out within this latent space. This method enables us to regulate the dimensionality of the atomic and bond embeddings, which directly impacts the overall number of model parameters. Given that atomic and bond vectors are typically sparse and limited to 118 distinct atomic vectors (with slight variations based on atomic environments), encoding these vectors results in minimal loss of information. This encoding approach allows for precise control over model complexity, which plays a crucial role in determining training convergence and mitigating the risk of overfitting.

2.3 Multi-Graph Representation of a Molecule

A natural way to represent molecular information is through graphs, resulting in a multi-graph representation that fully encodes the previously mentioned data. This structure consists of three distinct graphs: (a) an atom-bond graph, (b) a bond-angle graph, and (c) an angle-dihedral angle graph. In the atom-bond graph, atoms are represented as nodes, with bonds forming the edges between them. Similarly, the bond-angle graph represents bonds as nodes, with edges capturing the angles between pairs of adjacent bonds. The angle-dihedral angle graph follows the same principle, with dihedral angles connecting angles.

An important observation is that the bond-angle graph is the line graph of the atom-bond graph. In other words, the edges of the atom-bond graph become the nodes of the bond-angle graph, and adjacent edges in the atom-bond graph correspond to edges

Table 1: Atom embeddings

Property Name	Encoding Type	Sample Value
<i>Atom Type</i>	One Hot	'C', 'N', 'O', 'S', 'F', ...
<i>Number of Heavy Atoms</i>	One Hot	0, 1, 2, 3, 4, "More Than Four"
<i>Formal Charge</i>	One Hot	-3, -2, -1, 0, 1, 2, 3, "Extreme"
<i>Hybridization</i>	One Hot	"S", "SP", "SP2", "SP3", "SP3D", "SP3D2", "OTHER"
<i>Is in Ring?</i>	Label	0, 1
<i>Is aromatic?</i>	Label	0, 1
<i>Normalized Atomic Mass</i>	Scaler	-
<i>Normalized Van der Waal's Radius</i>	Scaler	-
<i>Normalized Covalent Radius</i>	Scaler	-
<i>Chirality Type</i>	One Hot	"CHI_UNSPECIFIED", "CHI_TETRAHEDRAL_CW", "CHI_TETRAHEDRAL_CCW", "CHI_OTHER"
<i>Number of Hydrogens</i>	One Hot	0, 1, 2, 3, 4, "More Than Four"

Table 2: Bond embeddings

Property Name	Encoding Type	Sample Value
<i>Bond Type</i>	One Hot	SINGLE, DOUBLE, TRIPLE, AROMATIC
<i>Is in Conjugated Bonds?</i>	Scaler	0, 1
<i>Is in ring?</i>	Scaler	0, 1
<i>Stereo Type</i>	One Hot	"STEREOZ", "STEREOE", "STEREOANY", "STEREONONE"

in the bond-angle graph. This relationship also holds between the bond-angle graph and the angle-dihedral angle graph.

These shared nodes serve as links between the three graphs, facilitating the propagation of information across them by treating these nodes as common points. This inter-graph information flow forms the foundation of the message-passing mechanism, which is further detailed in Section 2.5.

2.4 Neural Message Passing on Graphs

In this section, we discuss how we apply neural message passing on all 3 graphs of the molecular representation and pass information between these graphs. We first discuss neural message passing on a single graph.

The key idea behind neural message passing is that of the *message pass* iteration. In a message pass iteration, every node sends information about itself to all of its neighbors. Upon receiving this information, every neighbor updates their own internal state. After this process is repeated for a fixed number of iterations T , every node has information about itself and its T^{th} degree neighbors. Let $G = (V, E)$ be a graph, let $h_u^{(k)}$ denote the hidden state of node u at timestep k and let $x_{uv}^{(k)}$ denote the same for a edge originating at node u and terminating at node v

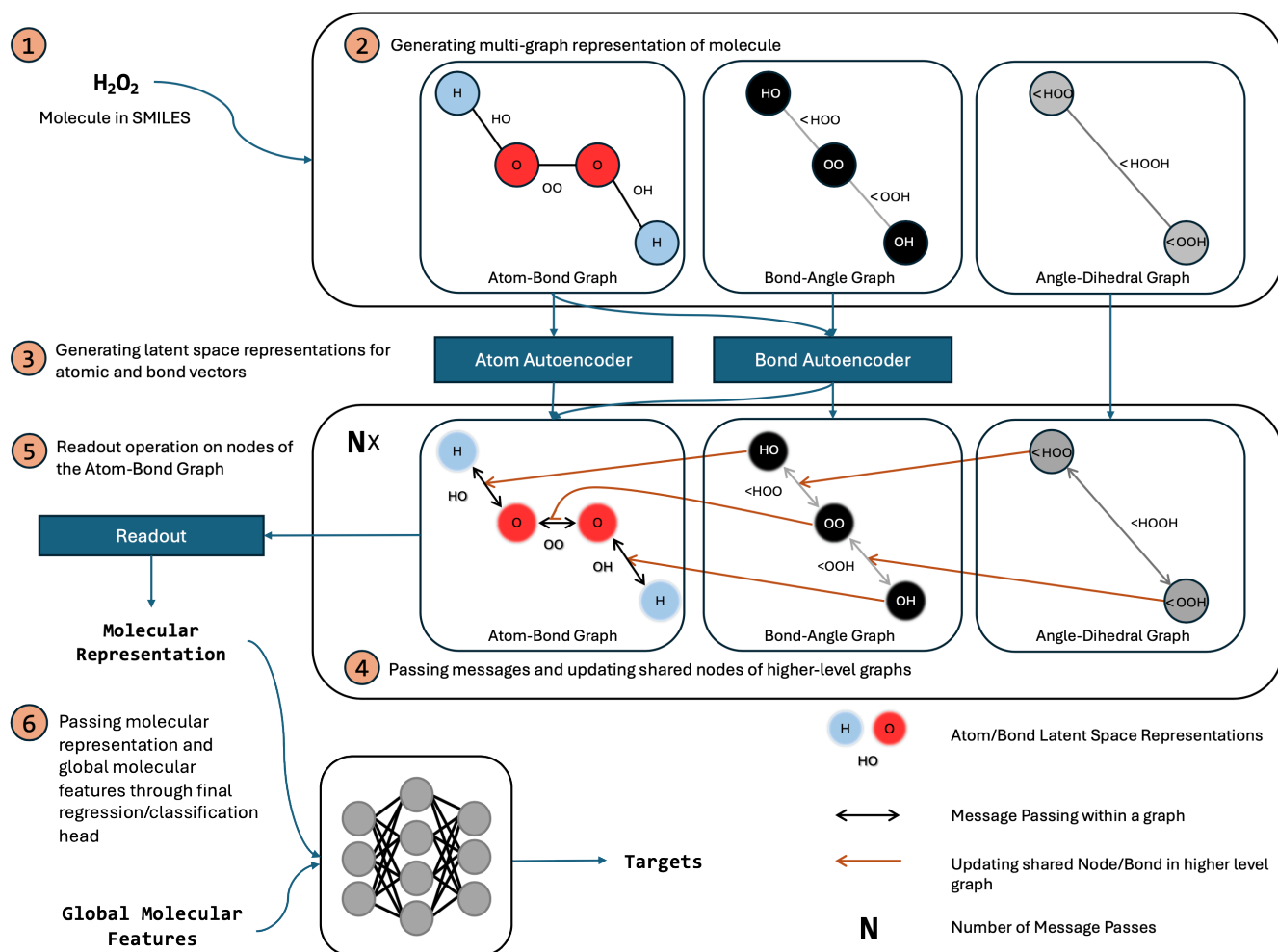


Figure 1: **Overall architecture:** Steps (1) and (2) involve generating initial features for the model, specifically, three graphs that capture the full 3D structure of the molecule along with relevant chemical properties. Step (3) generates latent space representations for appropriate node and edge features derived in step (2). In step (4), directed message passing is conducted on each graph within the molecule's representation. Step (5) performs a readout operation, aggregating node features from the atom-bond graph after message passing. Finally, in step (6), the aggregated molecular features are passed through a regression/classification head to predict the target values or class labels. We set the tunable parameter for the number of message-passing iterations, N , to 3.

then a message pass iteration can be expressed as the following,

$$a_u^{(k+1)} = \text{AGGREGATE}^{(k+1)} \{m(h_u^{(k)}, h_v^{(k)}, x_{uv}^{(k)}) | v \in \mathcal{N}(u)\}$$

$$h_u^{(k+1)} = \text{COMBINE}^{(k+1)}(h_u^{(k)}, a_u^{(k+1)})$$

Here, m is the message function which transmits a message from v to u across uv . The aggregate operation combines all these messages to give a_u which is then given to the combine operation to update the node state. The aggregate operation is typically commutative. This process is typically iterated for multiple rounds to allow information to propagate through the graph.

We use a variation of original message passing called directed message passing, in this we maintain two different edges (from v to u and from u to v) that begin with the same initial representation but may change later on. Then we perform message passing as normal. Directed message passing has been found to perform better^[40].

2.5 Passing Information Between Graphs

We first establish the sequence of information flow across the graphs. Information is passed from the lowest-level graph, the angle-dihedral graph, to the intermediate bond-angle graph. Subsequently, the information from the bond-angle graph is passed to the atom-bond graph, where a readout is performed to generate the final molecular representation.

The specific order of message passing can follow two approaches: (a) complete message passing within a lower-level graph before transferring shared nodes to the higher-level graph, or (b) perform a single message pass in the lower-level graph, then transfer information to the higher-level graph, where message passing continues.

We adopt the latter approach, as it enables more frequent exchanges of information between graphs at each stage, while maintaining similar computational cost.

2.6 Classification/Regression

Once message passing is complete, we obtain updated embeddings for each atom. These embeddings are subsequently combined using a readout mechanism, which in our case is implemented as an attention-based fusion^[41] strategy. The resulting molecular embedding is concatenated with 200 global molecular features generated using descriptastorous^[42,43] library.

This combined vector is then passed through a feedforward neural network for the final regression or classification task. The size of the output layer is determined by the number of regression targets or the number of classes in classification tasks. For classification, we apply a softmax function to the output logits.

2.7 Benchmarks

We compared performance of our GNN3Dihed with several molecular property prediction state-of-the-art GNN/MPNN models, such as, D-MPNN^[40], AttentiveFP^[44], SAMPN^[45], EGNN^[38], AGBT^[46], TIDAL^[47], and GEM^[24].

2.8 Datasets

Solubility regression data: FreeSolv, ESOL, LogS, and Lipo, and toxicity classification data: BACE, BBBP, ClinTox, ToxCast, and SIDER were taken from MoleculeNet^[48]. Other datasets were mu^[47], GPMT^[49], Ames^[50], hERG and hDAT^[51] LD50, IGC50, LC50, and LC50DM^[52], and QM9^[53,54]. For more information about datasets, refer Supporting Information. We experimented on multiple molecular benchmarks for both regression and classification tasks. We used random-splits with the ratio of 90:10 for train/test sets; further, the train set was split into a train and validation by 90:10. For each dataset the test set is a holdout set.

3 Results

We evaluated the performance of our model on the test set of various datasets. The mean and standard deviation of the performance metric were computed from the six independent runs. Table (3) shows performance of our model on various datasets. We have used our standard GNN3Dihed model consistently throughout the manuscript, unless otherwise indicated. A GNN3Dihed model by default uses the multi-graph formalism. Our models were trained on single target for both regression and classification tasks.

The root mean squared error (RMSE) was computed to evaluate the performance of each regression task. Whereas for the classification model each target was associated with the binary label therefore, standard binary cross entropy was used as loss function to optimize the model. We used AUC-ROC (area under the receiver operating characteristic curve) as the evaluation metric.

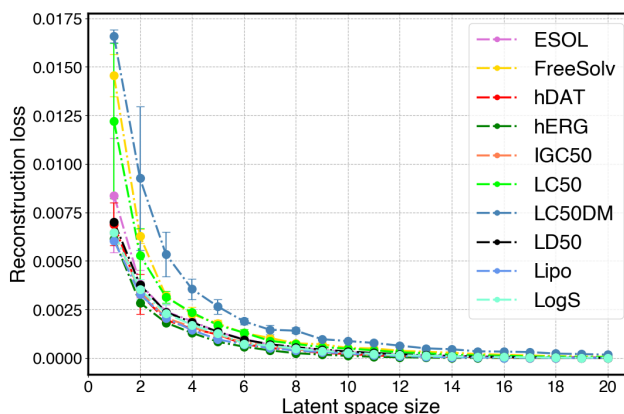
3.1 Determination of Latent Space Sizes

The optimal latent space size results from a tradeoff between minimizing reconstruction loss and reducing the latent space dimension. Typically, the goal is to select the smallest latent space that maintains a low reconstruction loss. To determine the latent space size, we evaluated the reconstruction loss of the atom and bond autoencoders on 10 regression datasets across different latent space sizes. For each size, we sampled the reconstruction loss five times, and the corresponding mean values

and standard deviations were calculated. They are provided in Figures 2(a) and 2(b). For both atom and bond autoencoders it can be seen that how reconstruction loss is reducing as latent space size is increasing. Eventually, reconstruction loss becomes negligible at latent space size ≥ 10 and ≥ 4 for atom autoencoder and bond autoencoder, respectively.

Our analyses have shown that only using the atom-bond graph, with optimal latent space, we can achieve a regression loss (RMSE) of 0.925 compared to the loss, without latent space representations, which stands at 2.082 (see Table 3), on the FreeSolv dataset. This is a significant improvement in accuracy considering that we are not using angle or dihedral angle information. Moreover, this improvement in accuracy comes with a minimal cost to the performance, as predictions are made only using the atom-bond graph. Thus, the usage of latent space representations lead to significant improvements in accuracy with minimal overhead.

(a)



(b)

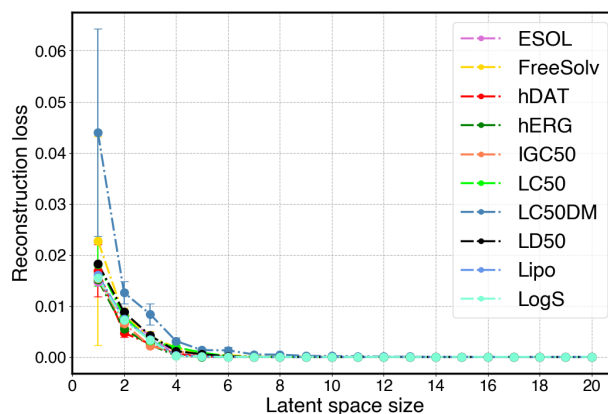


Figure 2: **Reconstruction loss with increasing latent space size:** (a) atom autoencoders, and (b) bond autoencoders for 10 regression datasets. Standard deviations from five independent runs are shown as error bars.

Table 3: Comparison of the performance of GNN3Dihed with the state-of-the-art models using ten regression and ten classification datasets

Regression (lower the better ↓); evaluation metric RMSE									
Models	FreeSolv	ESOL	Lipo	LD50	IGC50	LC50	LC50DM	hERG	hDAT
D-MPNN	2.082	1.050	0.683	-	-	-	-	-	-
AttentiveFP	2.073	0.877	0.721	-	-	-	-	-	-
EGNN	-	-	-	0.660	-	-	-	-	-
AGBT	0.994	-	0.570	0.554	0.391	0.692	0.743	-	-
TIDAL	-	-	-	-	-	-	-	0.568	0.663
GEM	1.877	0.798	0.660	-	-	-	-	-	-
GNN3Dihed	0.961 _(0.136)	0.496 _(0.036)	0.595 _(0.023)	0.481 _(0.009)	0.422 _(0.026)	0.717 _(0.036)	0.880 _(0.060)	0.615 _(0.020) ^a	0.718 _(0.022) ^b
LogS / RMSE: 0.535 _(0.037) using GNN3Dihed; 0.661 using SAMPN ^[45] ; ^a 0.64 using QSAR models ^[55] ; ^b 0.71 using QSAR models ^[55]									
Classification (higher the better ↑); evaluation metric ROC-AUC									
Models	BACE	BBBP	ClinTox	ToxCast	SIDER	Ames	mu	hERG	hDAT
D-MPNN	0.809	0.710	0.906	0.655	0.570	-	-	-	-
AttentiveFP	0.784	0.643	0.847	0.637	0.606	0.814	-	0.825	-
EGNN	0.832	0.960	0.851	0.685	0.606	0.836	-	0.839	-
TIDAL	-	-	-	-	-	-	0.987	0.988	0.989
GEM	0.856	0.724	0.901	0.692	0.672	-	-	-	-
GNN3Dihed	0.817 _(0.008)	0.888 _(0.003)	0.722 _(0.031)	0.515 _(0.012)	0.661 _(0.011)	0.825 _(0.003)	0.922 _(0.003)	0.944 _(0.003)	0.923 _(0.034)
GPMT / AUC: 0.729 _(0.011) using GNN3Dihed; 0.688–0.764 using various QSAR models ^[49]									

3.2 Regression Benchmarks

Before going into producing performance metrics for all datasets, we checked the ability to learn data by our model. Therefore, we calculated learning curves for 10 regression datasets, given in Figure 3. Here, prediction losses (RMSEs) are plotted against increasing training set size. This figure shows that GNN3Dihed is learning the data quite well as the RMSE is systematically decreasing with increasing number of samples in the training set.

The overall performance of our GNN3Dihed is given in Table 3, and compared with state-of-the-art various GNN architectures where molecular geometry information is incorporated. For datasets FreeSolv, ESOL and Lipo, our model outperforms GEM which is the current state-of-the-art model to include geometry information. However, GEM considers atom-bond and bond-angle graphs. The AGBT (algebraic graph-assisted bidirectional transformer) model captures 3D information of a molecule by forming weighted colored element-specific sub-graph representations^[46]. The TIDAL (topology-inferred drug addiction learning), another contribution from Wei and co-workers^[47] makes use of 3D persistent Laplacian (PL) and deep bidirectional transformers. GNN3Dihed achieves ~3% and ~7% improvement over AGBT in case of FreeSolv and LD50, respectively. However, AGBT has ~3–5% better performance in case of Lipo, IGC50, LC50 hERG and hDAT datasets. Surprisingly, LC50DM has ~14% lower RMSE in AGBT than in our model. The LC50DM dataset, the smallest of all 22 datasets studied presently has only 353 molecules, 316 were used for training and evaluation whereas 36 were used as test set (one molecule skipped, see Supporting Information for more details). In case of smaller dataset ML models tend to overfit if a large number of features are used. Our model seems to learn effectively on smaller datasets, for example, FreeSolv, LC50DM, and hDAT satisfactorily well (see Figure 3). This could be due to the use of autoencoders, which generate highly compressed latent space representations. This compression reduces the number of parameters, enabling the model to fit effectively even when working with limited data. GEM shows RMSE of 1.877 for FreeSolv whereas our model shows 0.961 (see Table 3). This is the best performance reported till date for the FreeSolv dataset. Our

model, includes torsional angles in the right form in comparison to the best reported molecular geometry enhanced model GEM where torsional angles were not part of the GNN, achieves SOTA results, overall.

Additionally, performance of GNN3Dihed on QM9 dataset, the largest among those used in this work, is checked. The HOMO-LUMO gap (hartree unit) was chosen as the molecular property. The evaluation metric mean absolute error (MAE) is used to measure the prediction accuracy. Our model gives MAE of 0.013 whereas GEM predicts MAE of 0.00746^[24] using 12 tasks.

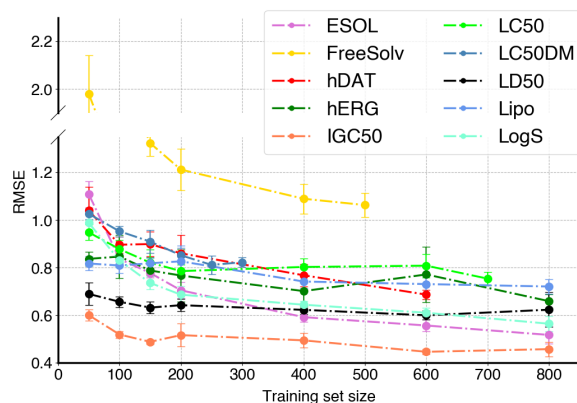


Figure 3: **Learning curves:** Prediction accuracies estimated by an evaluation metric, RMSE as a function of the training set size for 10 regression datasets. Standard deviations from five independent runs are shown as error bars. For generating learning curves we have considered largest training set size of 800 whereas, for smaller datasets, for example, FreeSolv had 500, hDAT had 600, LC50 had 700, and LC50DM had 300 training samples in the largest training set.

3.3 Classification Benchmarks

The performance metric, mean AUC and its standard deviation on 9 classification datasets are given in Table 3. Performances of classification tasks on ToxCast and SIDER are heavily influenced by the distribution of binary labels (see Supporting Information for details) for all of the baselines and our model too. ToxCast suffers from high class-imbalance. We can see the least AUC of 0.515. This dataset has shown lower AUC consis-

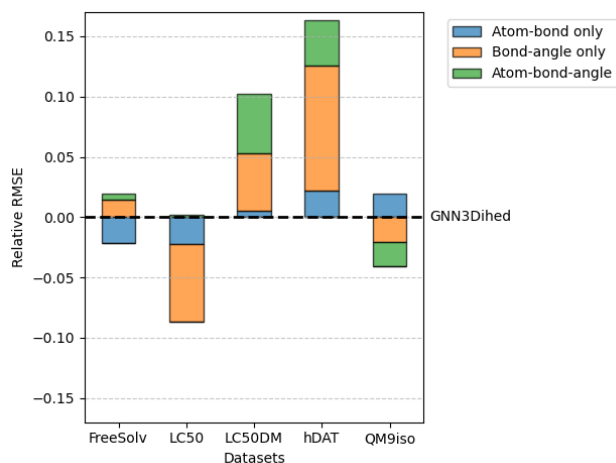


Figure 4: **Relative performance of multi-graph variants:** Prediction accuracies (RMSE) of atom-bond only, bond-angle only, and atom-bond-angle models with respect to the GNN3Dihed.

tently throughout all SOTA models. GEM was shown to give better AUC of 0.901 for ClinTox even though ClinTox as well suffers from class-imbalance. Contrary to ClinTox, ToxCast and SIDER, the rest of the classification datasets has well balanced class labels. Our model performs on BACE, BBBP, Ames, mu, hERG, and hDAT with an AUC of > 0.8 . The best performance of our model was seen on hERG dataset with an AUC of 0.944 which is 4.4% lower than the TIDAL model. TIDAL has better AUC of $\sim 5\text{--}7\%$ on hERG, hDAT, and mu compared to our model. Essentially, our model achieves the same AUC as EGNN model on Ames dataset. GNN3Dihed and EGNN have AUC values of 0.827 and 0.836, respectively. Our model has about 16% higher AUC value than GEM on BBBP dataset. However, EGNN outperforms our model by $\sim 7\%$ high AUC value on the same dataset. It is apparent from Table 3 that AUC values obtained from our GNN3Dihed model achieved results that are on par with SOTA models.

3.4 Effect of Geometry

To investigate the effect of geometry of a molecule we checked the performance of our model with/without some graphs. One of the important aspects of our architecture is flexibility on graph model (*e.g.*, atom-bond, bond-angle etc.) selection. Figure 4 shows performance (evaluation metric RMSE) of atom-bond, bond-angle, and atom-bond-angle graphs relative to the full atom-bond-angle-dihedral graph which is nothing but our standard GNN3Dihed. For this experiment, we chose three smaller regression datasets, FreeSolv, LC50, and LC50DM, and a larger regression dataset QM9iso which is consisted of constitutional isomers (see Supporting Information for details on

stoichiometries used) extracted from the QM9 dataset. FreeSolv and QM9iso have similar contributions coming from atom-bond, bond-angle, and atom-bond-angle graphs relative to the GNN3Dihed model; a cumulative variation of $\pm 2.5\%$. In case of LC50, atom-bond and bond-angle graphs have lower RMSEs by about 2.5% and 6.5%, respectively. However, for the LC50DM dataset GNN3Dihed achieves improvement cumulatively by 10% from various graph components. In certain datasets, atom-bond graph had shown to have lower RMSEs than the full atom-bond-angle-dihedral graph. This indicates some datasets are more sensitive to certain features^[46].

To gain more insights into the geometry aspects on the model performance we looked into the correlation data between predictions and targets. This experiment will ensure how an individual molecule's prediction differs from the target. QM9iso regression dataset was chosen for further analysis. The Pearson correlation (PC) coefficients were compared in Figures 5a and 5b after removing a single outlier from both atom-bond and GNN3Dihed models. The later achieved a larger PC coefficient of 0.7253, and atom-bond model had 0.7050. In Figure 5c, five molecules are shown with least prediction accuracies. GNN3Dihed consistently gave better RMSE values compared to atom-bond model. In Figure 5d, five molecules are shown with best prediction accuracies for atom-bond model. It is interesting to note that GNN3Dihed gives RMSE values close to atom-bond model. Comparing both cases it is clear that GNN3Dihed is more suitable for property predictions of chemical structures of diverse classes.

Another experiment was performed to further corroborate geometry effect. For this mu classification dataset was chosen. GNN3Dihed model has performance metric AUC of 0.922. Here, the holdout test set contains 300 samples from which 171 samples were filtered out based on the 5-membered heterocycles as they are known to influence drug related toxicity assays^[56,57]. Among them 10.5% samples were misclassified by our model. From the correctly predicted samples, prediction probability of two samples including their structures are given in Figure 6. Among these two samples, the one containing thiophene heterocycle is an active compound towards drug addition, whereas the other one with an amide moiety shows no activity. Our model correctly classifies them with 0.72 and 0.25 probabilities.

Conclusions

In principle the graph objects nodes and edges are suitable for representing atoms and bonds of a chemical structure. However computing accurate properties of a molecule often require a full consideration of the 3D geometry information. Additionally, the challenge is how to efficiently encode these information via various graph objects. In the current work, we have proposed a general strategy to systematically encode full 3D geometry information, like, atoms, bonds, angles, and dihedral angles using multi-graph formalism. Additionally, we have employed latent space embeddings derived from autoencoders in the neural message passing stage. The performance of the current GNN3Dihed model was compared with the existing various state-of-the-art baselines on several tasks (regression and classification), *e.g.*, solubility prediction, toxicity prediction, binding affinity, and quantum mechanical property prediction—cumulatively on 22 datasets. Our model showed some of the best molecular property predictions till date. Besides, our model was tested on the QM9iso—a dataset consisting of constitutional isomers in order

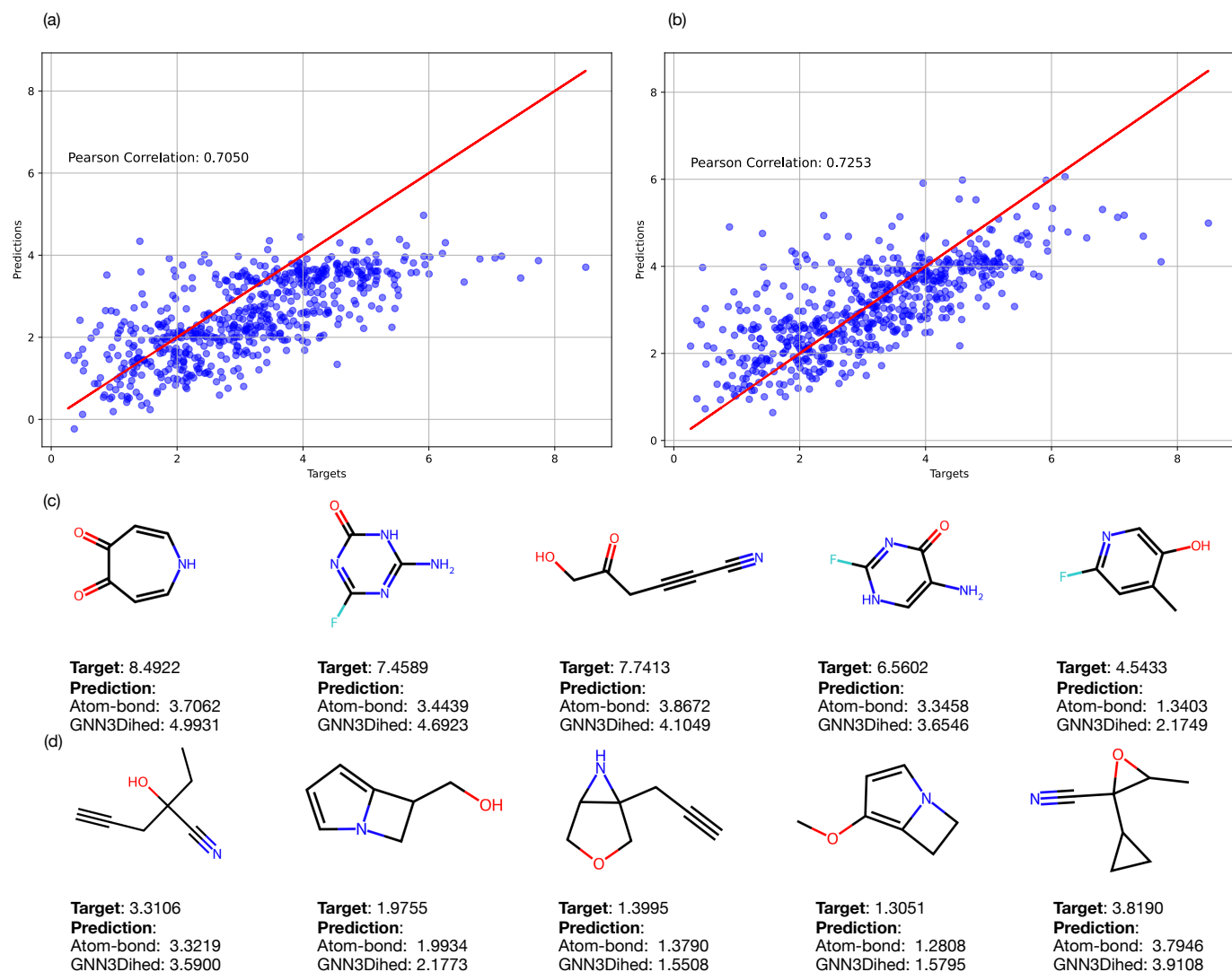


Figure 5: **Prediction on QM9iso data:** (a) atom-bond (b) GNN3Dihed models' Pearson correlation coefficients, (c) selected structures (five out of top-ten worst predicted by atom-bond) with their target properties, dipole moments in Debye units, and (d) selected structures (five out of top-ten best predicted by atom-bond) with their target properties, dipole moments in Debye units

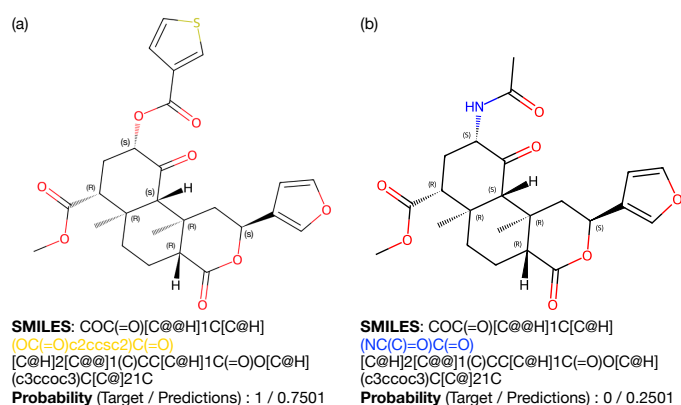


Figure 6: **5-membered heterocycles from mu dataset:** Chemical structures with their corresponding SMILES strings, and target and prediction probabilities from classification model

to check performances of variants of multi-graph models. This test indicated that the usage of information related to the full 3D geometry of a molecule improved the prediction accuracy as well as the flexibility of our model architecture.

Author contributions

SARS: Coding; formal analysis; writing – original draft, review and editing. SR: Conceptualization and supervision; coding, data curation; formal analysis; writing – original draft, review and editing.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

We thank HPC facilities of Department of Computer Science and Engineering, Mahindra University.

References

- [1] Kevin P Murphy. *Machine learning: a probabilistic perspective*. MIT press, 2012.
- [2] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *Nature*, 521(7553):436–444, 2015.
- [3] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT press, 2016.
- [4] Mati Karelson, Victor S Lobanov, and Alan R Katritzky. Quantum-chemical descriptors in QSAR/QSPR studies. *Chem. Rev.*, 96(3):1027–1044, 1996.
- [5] Jie Shen and Christos A Nicolaou. Molecular property prediction: recent trends in the era of artificial intelligence. *Drug Discovery Today: Technologies*, 32:29–36, 2019.
- [6] Oliver Wieder, Stefan Kohlbacher, Méline Kuenemann, Arthur Garon, Pierre Ducrot, Thomas Seidel, and Thierry Langer. A compact review of molecular property prediction with graph neural networks. *Drug Discovery Today: Technologies*, 37:1–12, 2020.
- [7] Matthias Rupp, Alexandre Tkatchenko, Klaus-Robert Müller, and O Anatole von Lilienfeld. Fast and accurate modeling of molecular atomization energies with machine learning. *Phys. Rev. Lett.*, 108(5):058301, 2012.
- [8] Katja Hansen, Franziska Biegler, Raghunathan Ramakrishnan, Wiktor Pronobis, O Anatole von Lilienfeld, Klaus-Robert Müller, and Alexandre Tkatchenko. Machine learning predictions of molecular properties: Accurate many-body potentials and nonlocality in chemical space. *J. Phys. Chem. Lett.*, 6(12):2326–2331, 2015.
- [9] Felix A Faber, Anders S Christensen, Bing Huang, and O Anatole von Lilienfeld. Alchemical and structural distribution based representation for universal quantum machine learning. *J. Chem. Phys.*, 148(24):241717, 2018.
- [10] Anders S. Christensen, Lars A. Bratholm, Felix A. Faber, and O. Anatole von Lilienfeld. FCHL revisited: Faster and more accurate quantum machine learning. *J. Chem. Phys.*, 152(4):044107, 2020.
- [11] Albert P Bartók, Risi Kondor, and Gábor Csányi. On representing chemical environments. *Phys. Rev. B*, 87(18):184115, 2013.
- [12] Oliver T Unke and Markus Meuwly. A reactive, scalable, and transferable model for molecular energies from a neural network approach based on local information. *J. Chem. Phys.*, 148(24):241708, 2018.
- [13] Shampa Raghunathan and U. Deva Priyakumar. Molecular representations for machine learning applications in chemistry. *Int. J. Quantum Chem.*, 122(7):e26870, 2022.
- [14] David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *J. Chem. Inf. Model.*, 50(5):742–754, 2010.
- [15] Alexandru T Balaban. Applications of graph theory in chemistry. *J. Chem. Info. Comput. Sci.*, 25(3):334–343, 1985.
- [16] Shinsaku Fujita. Description of organic reactions based on imaginary transition structures. 1.Introduction of new concepts. *J. Chem. Info. Comput. Sci.*, 26(4):205–212, 1986.
- [17] Franco Scarselli, Marco Gori, Ah Chung Tsoi, Markus Hagenbuchner, and Gabriele Monfardini. The graph neural network model. *IEEE transactions on neural networks*, 20(1):61–80, 2008.
- [18] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1263–1272, 2017.
- [19] Johannes Gasteiger, Janek Groß, and Stephan Günnemann. Directional message passing for molecular graphs. In *International Conference on Learning Representations (ICLR)*, 2020.
- [20] Connor W Coley, Wengong Jin, Luke Rogers, Timothy F Jamison, Tommi S Jaakkola, William H Green, Regina Barzilay, and Klavs F Jensen. A graph-convolutional neural network model for the prediction of chemical reactivity. *Chemical Science*, 10(2):370–377, 2019.
- [21] Kristof Schütt, Pieter-Jan Kindermans, Huziel Enoc Sauceda Felix, Stefan Chmiela, Alexandre Tkatchenko, and Klaus-Robert Müller. SchNet: A continuous-filter convolutional neural network for modeling quantum interactions. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.
- [22] David K Duvenaud, Dougal Maclaurin, Jorge Iparraguirre, Rafael Bombarell, Timothy Hirzel, Alán Aspuru-Guzik, and Ryan P Adams. Convolutional networks on graphs for learning molecular fingerprints. In C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 28. Curran Associates, Inc., 2015.
- [23] Yasuhiro Yoshikai, Tadahaya Mizuno, Shumpei Nemoto, and Hiroyuki Kusuoka. Difficulty in chirality recognition for transformer architectures learning chemical structures from string representations. *Nat. Commun.*, 15(1):1197, 2024.

- [24] Xiaomin Fang, Lihang Liu, Jieqiong Lei, Donglong He, Shanzhuo Zhang, Jingbo Zhou, Fan Wang, Hua Wu, and Haifeng Wang. Geometry-enhanced molecular representation learning for property prediction. *Nature Machine Intelligence*, 4(2):127–134, 2022.
- [25] David Weininger. SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules. *J. Chem. Info. Comput. Sci.*, 28(1):31–36, 1988.
- [26] David Weininger, Arthur Weininger, and Joseph L Weininger. SMILES. 2. Algorithm for generation of unique SMILES notation. *J. Chem. Info. Comput. Sci.*, 29(2):97–101, 1989.
- [27] Emma King-Smith. Transfer learning for a foundational chemistry model. *Chem. Sci.*, pages 5143–5151, 2024.
- [28] Bharath Ramsundar, Peter Eastman, Patrick Walters, Vijay Pande, Karl Leswing, and Zhenqin Wu. *Deep Learning for the Life Sciences*. O’Reilly Media, 2019.
- [29] Esther Heid, Kevin P Greenman, Yunsie Chung, Shih-Cheng Li, David E Graff, Florence H Vermeire, Haoyang Wu, William H Green, and Charles J McGill. Chemprop: a machine learning package for chemical property prediction. *J. Chem. Inf. Model.*, 64(1):9–17, 2023.
- [30] KT Schütt, Pan Kessel, Michael Gastegger, KA Nicoli, Alexandre Tkatchenko, and K-R Müller. Schnetpack: A deep learning toolbox for atomistic systems. *J. Chem. Theory Comput.*, 15(1):448–455, 2018.
- [31] Kristof T Schütt, Oliver T Unke, and Michael Gastegger. Equivariant message passing for the prediction of tensorial properties and molecular spectra. In *Proceedings of the 38th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 9377–9388. PMLR, 2021.
- [32] Johannes Gasteiger, Chandan Yeshwanth, and Stephan Günnemann. Directional message passing on molecular graphs via synthetic coordinates. In *Advances in Neural Information Processing Systems 34th (NeurIPS 2021)*, volume 19 of *Advances in Neural Information Processing Systems*, pages 15421–15433, 2021.
- [33] Johannes Gasteiger, Florian Becker, and Stephan Günnemann. GemNet: Universal directional graph neural networks for molecules. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 6790–6802. Curran Associates, Inc., 2021.
- [34] Ilyes Batatia, David P Kovacs, Gregor Simm, Christoph Ortner, and Gábor Csányi. MACE: Higher order equivariant message passing neural networks for fast and accurate force fields. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 11423–11436. Curran Associates, Inc., 2022.
- [35] Zian Li, Xiyuan Wang, Yinan Huang, and Muhan Zhang. Is distance matrix enough for geometric deep learning? In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 37413–37447. Curran Associates, Inc., 2023.
- [36] Victor Garcia Satorras, Emiel Hoogetboom, and Max Welling. E(n) equivariant graph neural networks. In *International Conference on Machine Learning*, pages 9323–9332. PMLR, 2021.
- [37] Oliver Unke, Mihail Bogojeski, Michael Gastegger, Mario Geiger, Tess Smidt, and Klaus-Robert Müller. SE(3)-equivariant prediction of molecular wavefunctions and electronic densities. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 14434–14447. Curran Associates, Inc., 2021.
- [38] Julian Cremer, Leonardo Medrano Sandomas, Alexandre Tkatchenko, Djork-Arné Clevert, and Gianni De Fabritiis. Equivariant graph neural networks for toxicity prediction. *Chemical Research in Toxicology*, 36(10):1561–1573, 2023.
- [39] Frank Harary and Robert Z Norman. Some properties of line digraphs. *Rend. Circ. Mat. Palermo*, 9:161–168, 1960.
- [40] Kevin Yang, Kyle Swanson, Wengong Jin, Connor Coley, Philipp Eiden, Hua Gao, Angel Guzman-Perez, Timothy Hopper, Brian Kelley, Miriam Mathea, et al. Analyzing learned molecular representations for property prediction. *J. Chem. Inf. Model.*, 59(8):3370–3388, 2019.
- [41] Chiori Hori, Takaaki Hori, Teng-Yok Lee, Ziming Zhang, Bret Harsham, John R Hershey, Tim K Marks, and Kazuhiko Sumi. Attention-based multimodal fusion for video description. In *Proceedings of the IEEE international conference on computer vision*, pages 4193–4202, 2017.
- [42] Brian Kelley. *Descriptor computation(chemistry) and (optional) storage for machine learning*, 2019. <https://github.com/bp-kelley/descriptastorus> [Accessed: Whenever].
- [43] Kevin Yang, Kyle Swanson, Wengong Jin, Connor Coley, Philipp Eiden, Hua Gao, Angel Guzman-Perez, Timothy Hopper, Brian Kelley, Miriam Mathea, Andrew Palmer, Volker Settels, Tommi Jaakkola, Klavs Jensen, and Regina Barzilay. Analyzing learned molecular representations for property prediction. *J. Chem. Inf. Model.*, 59(8):3370–3388, 2019.
- [44] Zhaoping Xiong, Dingyan Wang, Xiaohong Liu, Feisheng Zhong, Xiaozhe Wan, Xutong Li, Zhaojun Li, Xiaomin Luo, Kaixian Chen, Hualiang Jiang, et al. Pushing the boundaries of molecular representation for drug discovery with the graph attention mechanism. *J. Med. Chem.*, 63(16): 8749–8760, 2019.
- [45] Bowen Tang, Skyler T Kramer, Meijuan Fang, Yingkun Qiu, Zhen Wu, and Dong Xu. A self-attention based message passing neural network for predicting molecular lipophilicity and aqueous solubility. *J Cheminf*, 12:1–9, 2020.
- [46] Dong Chen, Kaifu Gao, Duc Duy Nguyen, Xin Chen, Yi Jiang, Guo-Wei Wei, and Feng Pan. Algebraic graph-assisted bidirectional transformers for molecular property prediction. *Nat. Commun.*, 12(1):3521, 2021.

- [47] Zailiang Zhu, Bozheng Dou, Yukang Cao, Jian Jiang, Yueying Zhu, Dong Chen, Hongsong Feng, Jie Liu, Bengong Zhang, Tianshou Zhou, and Guo-Wei Wei. Tidal: Topology-inferred drug addiction learning. *J. Chem. Inf. Model.*, 63(5):1472–1489, 2023.
- [48] Zhenqin Wu, Bharath Ramsundar, Evan N Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S Pappu, Karl Leswing, and Vijay Pande. Moleculenet: a benchmark for molecular machine learning. *Chem. Sci.*, 9(2):513–530, 2018.
- [49] Haoqiang Wang, Zejun Huang, Shang Lou, Weihua Li, Guixia Liu, and Yun Tang. In silico prediction of skin sensitization for compounds via flexible evidence combination based on machine learning and dempster-shafer theory. *Chem. Res. Toxicol.*, 37:894–909, 2024.
- [50] Katja Hansen, Sebastian Mika, Timon Schroeter, Andreas Sutter, Antonius Ter Laak, Thomas Steger-Hartmann, Nikolaus Heinrich, and Klaus-Robert Muller. Benchmark data set for in silico prediction of ames mutagenicity. *J. Chem. Inf. Model.*, 49(9):2077–2081, 2009.
- [51] Zailiang Zhu, Bozheng Dou, Yukang Cao, Jian Jiang, Yueying Zhu, Dong Chen, Hongsong Feng, Jie Liu, Bengong Zhang, Tianshou Zhou, et al. Tidal: topology-inferred drug addiction learning. *J. Chem. Inf. Model.*, 63(5):1472–1489, 2023.
- [52] Kedi Wu and Guo-Wei Wei. Quantitative toxicity prediction using topology based multitask deep neural networks. *J. Chem. Inf. Model.*, 58(2):520–531, 2018.
- [53] Raghunathan Ramakrishnan, Pavlo O Dral, Matthias Rupp, and O Anatole von Lilienfeld. Quantum chemistry structures and properties of 134 kilo molecules. *Sci. Data*, 1:140022, 2014.
- [54] Raghunathan Ramakrishnan. *A Python package for data-mining the QM9 dataset*, 2024. <https://github.com/raghurama123/qm9pack> [Accessed: Whenever].
- [55] Kuo Hao Lee, Andrew D Fant, Jiqing Guo, Andy Guan, Joslyn Jung, Mary Kudaibergenova, Williams E Miranda, Therese Ku, Jianjing Cao, Soren Wacker, et al. Toward reducing herg affinities for dat inhibitors with a combined machine learning and molecular modeling approach. *J. Chem. Inf. Model.*, 61(9):4266–4279, 2021.
- [56] Jeffrey B Sperry and Dennis L Wright. Furans, thiophenes and related heterocycles in drug discovery. *Current opinion in drug discovery & development*, 8(6):723, 2005.
- [57] Aura Rusu, Ioana-Maria Moga, Livia Uncu, and Gabriel Hancu. The role of five-membered heterocycles in the molecular structure of antibacterial drugs used in therapy. *Pharmaceutics*, 15(11):2554, 2023.